

AI-Trace Finder: Scanner Image Verification System

Milestone 4: Deployment & Final Documentation

Abstract:

- Identifying the source scanner or printer from scanned documents is a major challenge in digital forensics.
- Scanner source identification is crucial for document authentication, fraud detection, and forensic investigations.
- This project proposes an AI-based solution using deep learning techniques for scanner identification.
- The system aims to accurately classify scanned documents based on their originating scanner.

Problem Statement:

- Scanned documents from different scanners show very similar visual characteristics.
- Manual forensic analysis is time-consuming and requires expert knowledge.
- An automated intelligent solution is required for accurate scanner identification.

Objectives:

- Identify the scanner brand from scanned document images.
- Apply image preprocessing and feature extraction techniques.
- Develop a deep learning-based classification model.
- Evaluate the performance of the proposed system.

Dataset Description:

- Dataset consists of scanned document images collected from multiple scanners.
- Scanner brands include HP, Canon, Epson, and Brother.
- Each image is labeled with its scanner brand.
- Input is a scanned image and output is the predicted scanner brand.

Methodology:

- Collect scanned images under controlled conditions.
- Perform preprocessing such as resizing, normalization, and augmentation.
- Use CNN for automatic feature learning.
- Train and validate the model using labeled data.

System Architecture:

- Scanned document image is given as input.
- Image preprocessing enhances relevant features.
- CNN extracts scanner-specific patterns.
- System outputs the predicted scanner brand.

Model Implementation:

- CNN contains convolution, pooling, and fully connected layers.

- Model is trained with suitable epochs and batch size.
- Python is used for implementation.
- TensorFlow and OpenCV are used for development.

Results and Analysis:

- Accuracy and loss graphs are used for evaluation.
- Confusion matrix analyzes classification performance.
- Model correctly classifies HP, Canon, Epson, and Brother scanners.

Explainability:

- Grad-CAM highlights important regions used by the model.
- Scanner-specific texture and noise patterns are identified.
- Explainability increases trust in the system.

Technologies Used:

- Python
- TensorFlow
- OpenCV
- Jupyter Notebook
- Streamlit

Conclusion:

- AI-based scanner identification is effective and reliable.
- Deep learning improves accuracy over traditional methods.
- The system reduces manual forensic analysis effort.